

# Applied Analysis 7

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August 2024

## 1 Antibody activity of lupus patients in response to the influenza virus

Lupus is an autoimmune disease, which can affect any part of the body. As with other autoimmune diseases such as diabetes, the immune system attacks the body's own cells, causing inflammation and tissue damage. The full name *Systemic lupus erythematosus*, is abbreviated to **SLE**.

This question is based on data from ten lupus and 8 control patients who were given an influenza vaccine in tightly monitored circumstances. Lupus and control patients were selected at random from a pool of patients in a larger study in the U.S., and asked to take part in this influenza study. The measurements were made on those who gave informed consent; no information is available on the participation rate. The body responds to the vaccine by producing antibodies, a very large number of particles varying from one patient to another. The study aims to determine if the antibodies produced by SLE patients differ from those produced by the controls, and if so, in what ways. In immunology, an 'individual antibody' is not a single particle, but a monoclonal batch of particles identical at the molecular level. Each measurement on an individual antibody is a bulk measurement made on a monoclonal batch generated by a single cell extracted from the serum of one individual. Each antibody has an affinity value for the influenza virus, which varies from antibody to antibody and patient to patient.

The file `lupus.dat` contains information for 255 antibodies taken from the 18 patients. Of the 11 variables, the first five `sex`, `age`, `race`, `sle`, and `pid` (patient identity) are self-explanatory. The primary response `vaffinity` (antibody-virus affinity) is a measure of the antibody reaction to the influenza virus: this number is ordinarily in the range  $10^6 - 10^{10}$  in standard conventional units. For computational convenience the value given here has been re-scaled by multiplying by  $10^{10}$ . The influenza virus comes in several strains, and each antibody ordinarily reacts to a specific strain; `vaffinity` is the maximum of the strain-specific affinities. The strain to which each antibody reacts is recorded in `vstrain`, a 7-level factor, and the virus type in `vtype`, whose levels are **B**, **H1** and **H3**.

The number of antibodies per patient ranges from five to 43. One possibility is that these numbers are random as a consequence of laboratory protocols related to aliquot volumes and dilution levels selected by the experimenter independently of serum content. Another possibility is that antibody production levels vary randomly from patient to patient, and that these sample antibody numbers are roughly proportional to the individual profusions. Although the former scenario seems more plausible, some of the questions asked below are phrased as if the latter holds. In answering such questions, you should explain the implications of one scenario versus the other, and tailor your answers accordingly.

A certain fraction of the antibodies also react to another protein, called **RHA**. The active antibodies and their **RHA** affinity values are given in the variable `rhaff`, also multiplied by  $10^{10}$ .

Two further quantitative variables, the affinity values for two proteins **HA1** and **MN**, are also reported. Although formal statistical models may be useful to address some of the following questions, you should bear in mind that simple graphs and tables of averages may be equally compelling, especially if accompanied by a relevant standard error.

## Data Reading

```
lupus_data=read.table("lupus.dat.txt",header=TRUE)
head(lupus_data)
```

| ##   | sex | age | race | sle | pid  | vaffinity | rhaff    | HA1   | MN        | vtype | vstrain |
|------|-----|-----|------|-----|------|-----------|----------|-------|-----------|-------|---------|
| ## 1 | F   | 31  | C    | sle | SLE1 | 3100.00   | 0.791    | NA    | NA        | B     | MAL     |
| ## 2 | F   | 31  | C    | sle | SLE1 | 45.70     | 3.487    | NA    | NA        | H1    | SI      |
| ## 3 | F   | 31  | C    | sle | SLE1 | 209.00    | 28.390   | NA    | NA        | H1    | SI      |
| ## 4 | F   | 31  | C    | sle | SLE1 | 4130.00   | 2582.000 | NA    | NA        | H3    | WIS     |
| ## 5 | F   | 31  | C    | sle | SLE1 | 9.72      | 3.174    | 1.875 | 0.3703704 | B     | MAL     |
| ## 6 | F   | 43  | C    | sle | SLE2 | 1.57      | NA       | NA    | NA        | H3    | WIS     |

## 2 Possible Questions

### Problem 1

In terms of the descriptors commonly applied to scientific studies, comparative, randomized, replicated, case-control, and so on, how would you describe this study? Justify your answer. **True/False:** In terms of partitions `vtype` produces larger partitions compared to `vstrain`?

### Problem 2

Are the control patients appropriately selected? Do the controls differ systematically from the SLE patients in ways that are unrelated to the disease but could affect antibody activity comparisons?

### Problem 3

One group of key scientific questions is concerned with how and whether the antibodies produced by SLE patients are different from those produced by the controls, either in number of antibodies produced or in viral activity levels. `vaffinity` is the main variable that we are going to study for that analysis. What kind of data-preprocessing do you think is necessary before doing any inference?

### Problem 4

Do the SLE antibodies have greater or lesser viral affinity than the controls? Explain carefully what accommodations, if any, are made for the other variables sex, age etc.

### Problem 5

Is viral affinity related to the number of antibodies sampled? Compute a relevant test statistic with explanation as needed. How do you interpret this effect, if any?

## Problem 6

Do SLE patients produce antibodies in greater or lesser number than the controls? Explain your calculations and your reasoning.

## Problem 7

Not all antibodies react to the RHA protein. Let  $p_0$  be the fraction of control antibodies that exhibit RHA activity, and let  $p_1$  be the corresponding parameter for SLE patients. By using a Bayesian or other argument, compute the probability

$$\mathbb{P}(p_1 \geq p_0 | \text{data})$$

How will numerically compute it?

## Problem 8

The individual antibody affinity measurements are expensive and time-consuming. Nevertheless, the researcher is considering an extension of the experiment in order to pin down more accurately the difference in activity levels between SLE and control patients. The following proposals are similar in terms of cost:

1. Make 5 additional antibody measurements on each of the current patients, thereby enlarging the total number of measurements from 255 to 345.
2. Select four additional patients, two controls and two SLE, and make five antibody affinity measurements on each.

Assuming that all parameter values remain unchanged, which of the two scenarios lead to more reduction in the variance of the SLE effect?